# Minesweeper LLM Training with GRPO: Technical Summary

## 1. Executive Summary

This project trains a 14B parameter LLM to play Minesweeper using GRPO with LoRA, achieving 70% VRAM reduction. Three innovations: (1) multi-component reward function (R_total = R_format + R_gameplay) differentiating logical deduction (+15) from guessing (+10), (2) 10K-instance dataset with 50/50 CoT/non-CoT split for reasoning distillation (80% token reduction), (3) phased training from 6×6/8×8 to 25×25 grids. Current performance: 1% win rate, 50% JSON validity, 2.4 avg moves—demonstrating format learning but requiring extended training for strategic competency.

## 2. Reward Function Design

### 2.1 Architecture & Specification

The total reward, **R_total**, is calculated as the sum of a formatting reward, **R_format(a)**, and a gameplay reward, **R_gameplay(s,a,s')**.

**Reward Breakdown:**

| Event | Reward/Penalty |
|---|---|
| **Terminal** | |
| Win | +100 |
| Hit Mine | -25 |
| **Intermediate** | |
| Reveal Safe (Logical Deduction) | +15 |
| Reveal Safe (Guess) | +10 |
| Flag Correctly | +15 |
| Flag Incorrectly | -10 |
| Redundant Reveal | -12 |
| Redundant Flag | -8 |
| Invalid Move (Reveal) | -15 |
| Invalid Move (Flag) | -10 |

The system determines cell safety deterministically using constraint satisfaction, where a cell is safe if the mine count of an adjacent numbered cell is already satisfied by existing flags.

**Key Features and Advantages:**

- **Move Efficiency:** The reward function actively encourages the LLM to solve the Minesweeper board using the minimum number of clicks, utilizing the Bechtel board value as a core incentive.
- **Enhanced Feedback:** It provides dense intermediate feedback, offering a significant advantage over baseline models.
- **Logical Incentivization:** The structure specifically incentivizes logical deduction.

- **Graduated Penalties:** Penalties are graduated to accurately encode the severity of different mistakes.
- **Encourages Optimal Play:** The design is tailored to **encourage move efficiency** and optimal strategy.

## 3. Dataset & Reasoning Distillation

### 3.1 Composition & Hypothesis

10K manually curated instances across 5 tiers: 6×6 (5 mines), 10×10 (20), 15×15 (45), 20×20 (80), 25×25 (125). Stratified 50/50 partition: 5K with explicit CoT reasoning, 5K direct input→output. Hypothesis: CoT examples teach reasoning patterns; non-CoT examples force implicit application without verbalization. Benefits: reduced inference latency (20 vs 100+ tokens), concise output, enhanced generalization. Validation: automated state generation, constraint verification, manual CoT annotation, JSON schema compliance.

## 4. Phased Training Methodology

### 4.1 Phase 1: Constrained Bootstrapping

Exclusive 6×6/8×8 training, 50-100 steps, 15-20% mine density. Rationale: reduced state complexity, clearer feedback, faster convergence. Metrics: JSON validity 50%→90%, reward -0.15→+0.35, moves reduced to 2-4.

### 4.2 Phase 2: Knowledge Transfer

Introduced 10×10/15×15+ grids, 100-200 additional steps. Techniques: reduced learning rate, KL divergence monitoring (<0.014), gradient clipping, Phase 1 replay buffer. Hardware: AMD GPU, ROCm 7.0, 255GB memory, BFloat16. LoRA: 68.8M/14.8B parameters (0.46%), 2-6× speedup, ~4.5hr total training.

## 5. Results & Analysis

### 5.1 Performance Metrics (100-game eval)

**Win rate: 1%, Avg moves: 2.4, JSON validity: 50%, Completion length: ~20 tokens, Mean reward: -0.15 to +0.18, KL divergence: 0.001-0.014.** Analysis: Format learning successful; strategic reasoning underdeveloped. Premature termination via invalid actions suggests need for extended training.

### 5.2 Distillation & Phased Learning Impact

Distillation: 80% token reduction (20 vs 100+), partial CoT suppression observed. Phased training: JSON validity improved 50%→90% in 50 steps, knowledge transfer from 6×6/8×8 to 10×10 without catastrophic forgetting, accelerated convergence vs direct training on full complexity.

## 6. Conclusion & Future Work

### 6.1 Key Contributions

(1) Multi-component reward balancing safety, logic, efficiency; (2) 50/50 CoT distillation enabling implicit reasoning; (3) Curriculum phased training preventing catastrophic forgetting. Parameter-efficient LoRA (0.46% trainable) enables single-GPU training of 14B model.

### 6.2 Future Directions

Immediate: Extend to 500-1000 steps, add information-gain gradient rewards, scale dataset to 50K. Advanced: Hierarchical CoT distillation, self-play data generation, multi-step rollout evaluation, cross-domain transfer (Sudoku, logic puzzles), stress testing on 30×30+ grids and variable densities (10-40%).